



केन्द्रीय प्रदूषण नियंत्रण बोर्ड
CENTRAL POLLUTION CONTROL BOARD
पर्यावरण, वन एवं जलवायु परिवर्तन मंत्रालय भारत सरकार
MINISTRY OF ENVIRONMENT, FOREST & CLIMATE CHANGE GOVT. OF INDIA

Certificate No.: **CPCB/Mahaveer Polypack/Rajasthan/320**

Dated:- 01.03.2023

To,

M/s Mahaveer Polypack

G-150 Ajaymeru RIICO Industrial Area,
Palra, Ajmer - 305002

Sub: Provisional Certificate to manufacturer for Manufacturing / Selling of Compostable Carry Bags & Commodities.

With reference to the application no. **Mahaveer Polypack/Rajasthan/320**, Nil, dated **01 Mar 2023** this is to certify that **M/s Mahaveer Polypack** plant located at **G-150 Ajaymeru RIICO Industrial Area, Palra, Ajmer - 305002** is fulfilling the criteria as per revised Standard Operating Procedure (SOP) for issuing certificate as per the provisions '4(h)'&'11(c)' of Plastic Waste Management Rules, 2018, for manufacturing and selling of compostable carry bags and commodities in Indian Market as '**MANUFACTURER**'.

Certificate for manufacturing and selling of compostable plastic bags in Indian market is hereby issued to **M/s Mahaveer Polypack** plant located at **G-150 Ajaymeru RIICO Industrial Area, Palra, Ajmer - 305002** as '**MANUFACTURER**' with the following conditions:

- i. The end product "**Compostable plastics and commodities**" shall be manufactured using the raw materials "**PLA**" and following the production process (**Annexure-I**)
- ii. Each carrybags and commodities made from compostable material or plastic shall bear a label "**COMPOSTABLE**" **IS/ISO:17088** titled as Specifications for "Compostable Plastic" in **English & regional language**. Each carrybags and commodities shall also have printed code and Certificate Number of "**MANUFACTURER**" as given above.
- iii. The manufacturer shall generate QR code based on the details (Name, plant address, CPCB certificate no. etc.) provided in the certificate issued by CPCB and QR code shall be provided on each of the carry bags and commodities manufactured at the certified unit. The "verifiable" details of the QR code shall be shared with the SPCB/PCC/CPCB within one month of issue of this Certificate.
- iv. The Manufacturer shall provide six-monthly report giving details of raw material procurement and product sale to SPCB/PCC/CPCB as per the prescribed format.
- v. The "Manufacturer" shall comply with provisions of PWM Rules/ Guidelines issued from time to time by the Ministry of Environment, Forest & Climate Change or Central Pollution Control Board.

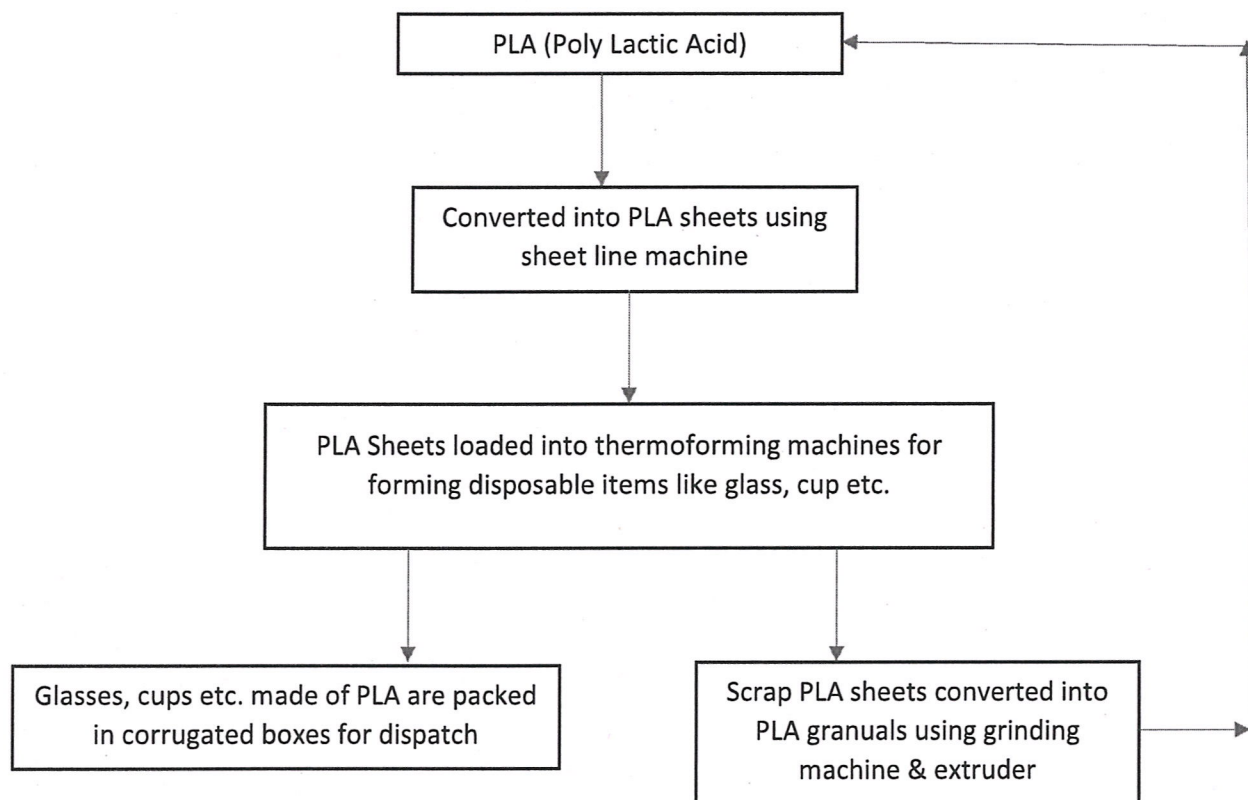
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- vi. If the certified Manufacturer is found non-complying with the provisions of the PWM Rules, 2018, the Certificate shall stand cancelled.
- vii. **The manufacturer shall submit a fresh complete test report of the sample within Six (6) months from the date of issue of Provisional Certificate as per Indian Standard IS/ISO:17088 (as amended time to time) from CIPET or any other CPCB recognized Government laboratory having adequate testing facilities for issuing final Certificate.**
- viii. The provisional certificate is valid for a period of six months from the date of issue of certificate by CPCB extendable by six months subject to the condition that the reasons for seeking extension as cited by the manufacturer are found valid by CPCB

दिव्या सिन्हा / Divya Sinha
वैज्ञानिक "ई" / Scientist "E"
(Divya Sinha)
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Central Pollution Control Board
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Parivesh Bhawan, East Arjun Nagar, Delhi-110 032

Manufacturing Flowchart of Compostable Plastic Items



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सिपेट : पेट्रोकेमिकल्स तकनीकी
संस्थान (आई.पी.टी.)

(पूर्व में केंद्रीय प्लास्टिक्स इंजीनियरिंग एवं तकनीकी संस्थान)

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प्लास्टिक परीक्षण केंद्र / PLASTIC TESTING CENTRE

श्रृंखला:ए / Series : **ANALYSIS REPORT** परीक्षण रिपोर्ट / TEST REPORT

क्र.सं / S.L. No. **12653**

को जारी / Issued to **MAHAVEER POLY PACK**
G-150, Ajaymeru RIICO Industrial
Area, 8, Palra, Ajmer - 306002,
Rajasthan, Mob.-8130930931

रिपोर्ट सं / Test Report No : **2596**

दिनांक / Dated : **27.01.2023**

संदर्भ / Your Ref. No. : **NIL**

दिनांक / Dated : **18.07.2022**

मानक के अनुसार टेस्ट रिपोर्ट
Test Report as per Standard :

As per part C

PART A

ULR No. TC816421000005365P

PARTICULARS OF SAMPLE SUBMITTED

- a) Name of the sample(as stated by the party) : Compostable Thermoforming Disposable Product
- b) Grade/Variety/Type/Size/Class etc. : ---
- c) Code No. : ---
- d) Quantity (pcs/mtr/gm/nos) : 1.5 kg.
- e) Mode of packing : In Poly Pouch
(Sealed carton/poly pouch/container or not)
- f) Date of receipt of sample : 18.07.2022
- g) Date of Performance of test : 22.07.2022
- h) Any other information : ---

PART B

SUPPLEMENTARY INFORMATION

- a) Reference to sampling procedure : Drawn & supplied by the party
- b) Supporting documents for Measurements : As per Part C
Taken and results derived like graphs,
Tables, sketches and/or Photographs as
appropriate to test report if any (to be attached)
- c) Deviation from the test methods as : Not Applicable
Prescribed in relevant ASTM/ISO/BIS/
Work instructions, if any


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ANALYSIS REPORT

**Report No.: 2596
Date : 27.01.2023**

**PART C
TEST RESULTS**

Sr. No.	Name of the Test	Test Method/Standard	Unit	Results obtained	Specified Requirements
Sample Details : Compostable Thermoforming Disposable Product (As stated by the party)					
1.	Material Identification	FTIR/DSC	--	Poly Lactic Acid (PLA)	-----
2.	Disintegration(Dry mass passing through 2 mm sieve after 84 days)	IS 17088:2008	%	8.8	Not more than 10%
3.	Ultimate aerobic Biodegradation (with reference to 100% degradation of positive reference)	IS 17088:2008	%	92	>90 (At the end of the test period not more than 180 days)
4.	Plant growth study Monocotyledon(Rice) % Seed emergence	IS 17088:2008	%	93.0	>90
	Dicotyledon(tomato) % Seed emergence		%	92.5	>90

Note: The detailed observation on biodegradability test is enclosed as Annexure -I


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प्लास्टिक परीक्षण केंद्र / PLASTIC TESTING CENTRE

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Report No.: 2596

Date : 27.01.2023

PART C
TEST RESULTS

Sr. no.	Name of the Test	Test Method/Standard	Unit	Results obtained	Specified requirements
5	Heavy metals concentrations # Arsenic (As) Copper (Cu) Nickel (Ni) Zinc (Zn) Cobalt (Co) Chromium (Cr) Molybdenum (Mo) Mercury (Hg) Cadmium (Cd) Lead (Pb) Selenium (Se)	IS 17088:2008	Mg/l	B.L.Q.(Q.L.=0.05) 0.81 0.70 B.L.Q.(Q.L.=0.05) B.L.Q.(Q.L.=0.05) 0.69 0.036 B.L.Q.(Q.L.=0.05) B.L.Q.(Q.L.=0.05) 0.13 B.L.Q.(Q.L.=0.05)	20 500 100 2500 ---- 300 ---- 10 20 500 ----

*Based on Municipal waste (Management and Handling) Rules, 1999 notified on 27th September, 1999 by Ministry of Environment and Forests, Government of India. Note that Concentration of Metals like Cobalt, Molybdenum, and Selenium is not mentioned in the notification.

PART D

REMARKS : Nil

Note :

1. This Test Report / Certificate is issued only for the samples submitted to the laboratory.
2. The results stated above related only to the items tested.
3. The quality of the subsequent production lot has to be ensured by the purchaser.
4. This Test Report shall not be reproduced except in full without the written approval of the laboratory.
5. Any anomaly/ discrepancy in this report should be brought to the notice of the laboratory within 30 days from the date of issue.
6. Subcontracted Tests (if any): # Gujarat Laboratory

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प्लास्टिक परीक्षण केंद्र / PLASTIC TESTING CENTRE

ANALYSIS REPORT

Report No.: 2596

Date : 27.01.2023

Annexure-I

OBSERVATION FOR BIODEGRADABILITY TEST AS PER IS /ISO 17088:2008

Customers Details:

Name of the Party :- MAHAVEER POLY PACK

1. Sample Details : : Compostable Thermoforming Disposable Product (As declared by the party)

2. Material Identification by FTIR & DSC: Polylactic Acid (PLA)

3. Observation:-

a) Conditions of reaction mixtures

Origin of Compost

: Livestock excreta, municipal and vegetable waste

Reaction Temperature

: 58° (±2°C)

Dry Solid

: 51.9 %

Volatile Solid

: 37.8 %

CO₂ evolved during 1st 10 days

in blank vessels

: 50.06 mg/g of volatile solids of compost

Test Duration

: 118 Days.

Reference Material

: Cellulose.

Volume of reaction vessel

: 3000ml.

b) pH of Test medium

Sl. No.	Composting Vessel	pH (before test)	pH (after test)
1	Blank 1	7.6	7.6
2	Blank 2	8.1	8.3
3	Blank 3	7.9	8.5
4	Cellulose 1	7.4	8.1
5	Cellulose 2	7.5	8.0
6	Cellulose 3	8.5	7.8
7	Negative 1	7.1	7.6
8	Negative 2	8.2	7.4
9	Negative 3	7.5	7.5
10	Sample 1	8.2	7.7
11	Sample 2	8.0	7.3
12	Sample 3	7.9	7.1

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संस्थान (आई.पी.टी.)

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ANALYSIS REPORT

Report No.: 2596

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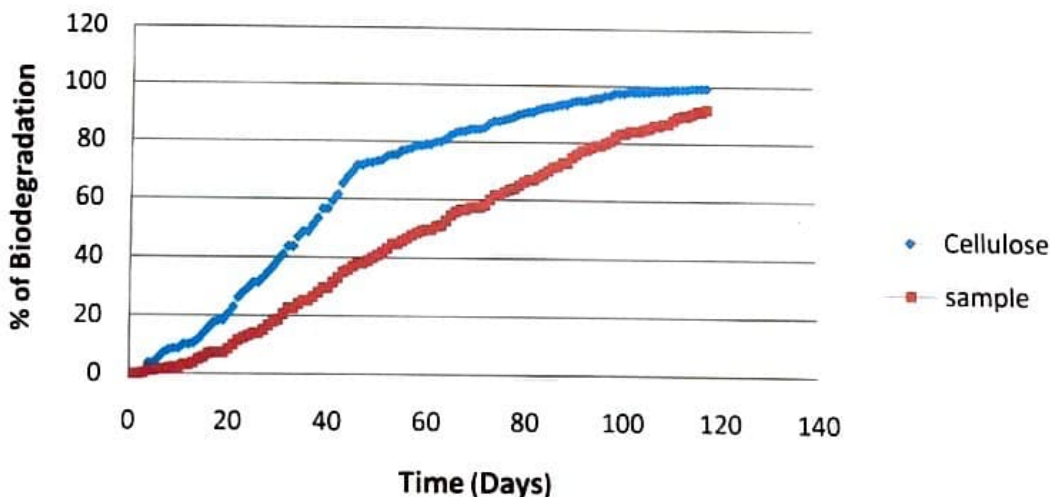
4. Result : Percentage biodegradation relative to positive reference

MEAN (%)

: 92 %

The reference material- Cellulose (%) : 100%

CODE No. 2074



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ANALYSIS REPORT

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5. Visual Observation of Sample

Description	Week 1	Week 5	Week 10	Week 15	Week 17
Structure	Glass sample	Glass sample	Glass sample	Glass sample	Glass sample
Moisture	Adequate Moisture level	Adequate Moisture level	Adequate Moisture level	Adequate Moisture level	Adequate Moisture level
Colour	Milky white	Milky white	Milky white	Faded white	---
Fungal Development	Nil	Nil	Nil	Nil	Nil
Smell	Organic/dirt like	Organic/dirt like	Organic/dirt like	Organic/dirt like	Organic/dirt like

6. Visual Observation of Compost

Description	Week 1	Week 5	Week 10	Week 15	Week 17
Structure	Fine Particles	Fine Particles	Fine Particles	Fine Particles	Fine Particles
Moisture	Adequate moisture level	Adequate moisture level	Adequate moisture level	Adequate moisture level	Adequate moisture level
Colour	Dark Brown	Dark Brown	Dark Brown	Dark Brown	Dark Brown
Fungal Development	Nil	Nil	Nil	Nil	Nil
Smell	Organic/dirt like	Organic/dirt like	Organic/dirt like	Organic/dirt like	Organic/dirt like


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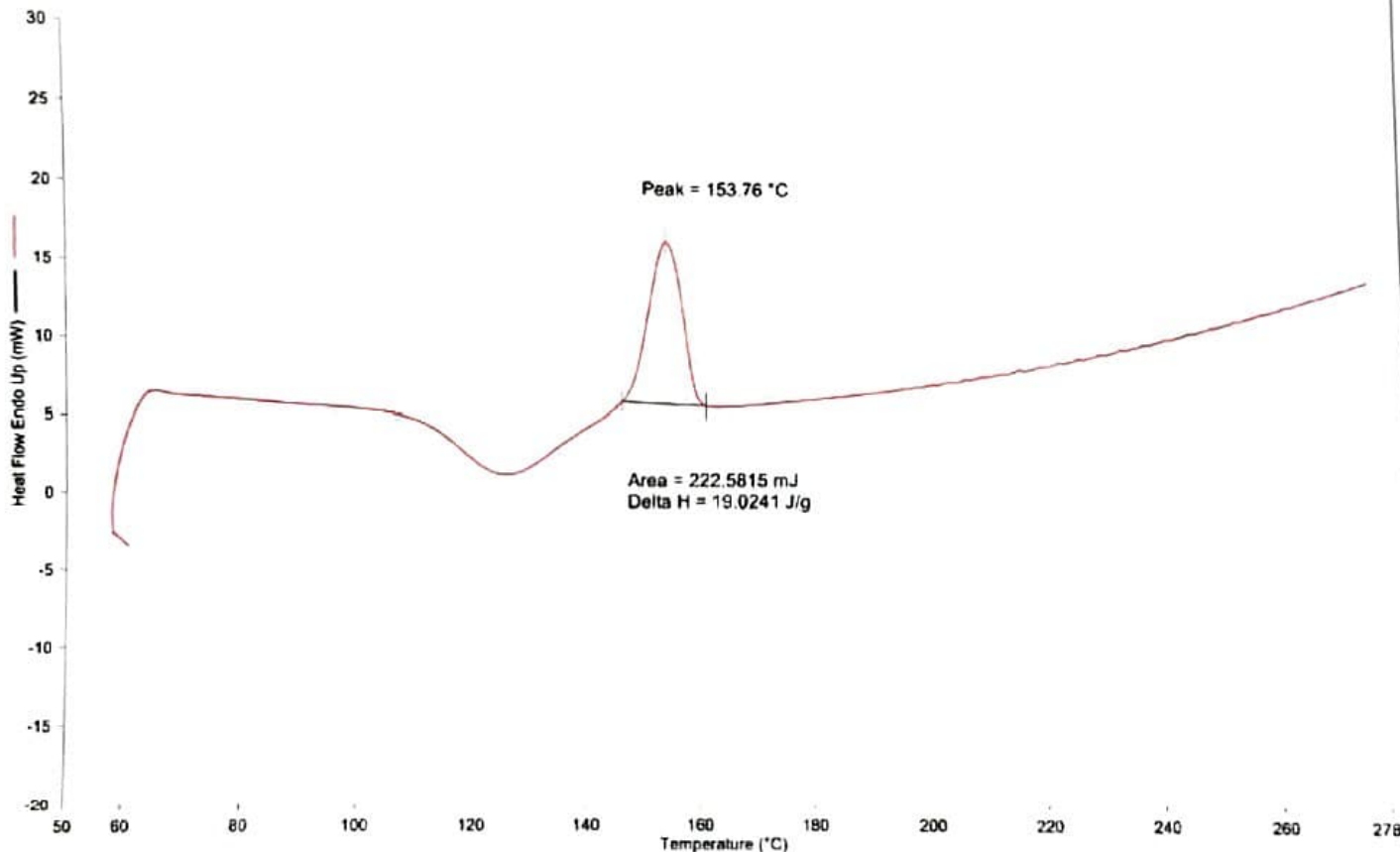
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8. DSC Analysis



Comment: The above DSC & FTIR analysis indicates the above sample is Polylactic Acid (PLA)
Compostable Thermoforming Disposable Product.

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पेट्रोकेमिकल्स तकनीकी
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9. DISINTEGRATION - AFTER 12 WEEKS

CODE NO : 2074

Before Disintegration



BEFORE DISINTEGRATION

CODE NO : 2074

After Disintegration



AFTER DISINTEGRATION

The disintegration of the sample by passing through 2 mm sieve after 12 week in composting condition as per ISO 17088-2012/IS 17088: 2008 was found not more than 10% of original dry mass remain.


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10. GERMINATION AND PLANT GROWTH STUDY



2074 Compost (Control)

2074 Compost (Sample)



2074 Compost (Control)

2074 Compost (Sample)

The percentage of seed germination rate if found to be greater than 90% for both Monocotyledon and Dicotyledon.

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INTERIM REPORT

IS 17899 T : 2022

ASSESSMENT OF BIODEGRADABILITY OF PLASTICS IN VARIED CONDITIONS (Aerobic)

ENVIRONMENTAL DIVISION LABORATORY, MUMBAI

INTERTEK INDIA PRIVATE LIMITED

IIPL/17025/ENV/QF/7.8/01-03	Issue No.: 01	Amend No.: 00
	Issue Date.: 16.12.2019	Amend Date.: 00.00.0000

Client: MAHAVEER POLYPACK

Sample Registration Date: 28/07/2022

Analysis starting date: 28/07/2022

Analysis completed on: in process

Name of product: BIODEGRADABLE AND COMPOSTABLE THERMOFORMING
PRODUCTS WITH 1% D2W

Quantity received and packing: - 500gms packing

Sample details: BIODEGRADABLE AND COMPOSTABLE THERMOFORMING PRODUCTS
WITH 1% D2W

Test Required: IS 17899 T: 2022 ASSESSMENT OF BIODEGRADABILITY OF PLASTICS
IN VARIED CONDITIONS (Aerobic)

Sampling done by: Sample not drawn by Intertek

Report No. MUM/002496A/2022

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LABORATORY

Testing as presented in this report was conducted by Environmental division of Intertek India Private Limited. The testing facility is located at F wing, 2nd Floor, Chandivali Saki vihar Road, Andheri (East), Mumbai – 400 072, India.

SAMPLE RECEIPT

The sample was received on 28/07/2022 at the Intertek testing facility. The sample was sent through courier. Sample was at ambient temperature in good condition with no evidence of damage or contamination. No temperature preservation was required.

SAMPLE DESCRIPTION:



Figure 1: BIODEGRADABLE AND COMPOSTABLE THERMOFORMING PRODUCTS

PROJECT DESCRIPTION:

BIODEGRADABLE AND COMPOSTABLE THERMOFORMING PRODUCTS WITH 1% D2W were submitted by MAHAVEER POLYPACK for testing under standard IS 17899T:2022. This Standard specifies the procedures and assessment of biodegradability of plastics under varied conditions. This provisional standard addresses the following aspects:

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1. Biodegradation
2. Negative effects of resulting biomass on terrestrial plant growth/organism.
3. Negative effects of the quality of the resulting biomass including the presence of high levels of regulated heavy metals.

This provisional standard is applicable for assessing the biodegradability of plastics under aerobic conditions.

Table 1 Assessment of Aerobic Biodegradability in Plastics under varied conditions

Sr. No.	Characteristics	Requirement	Method of Test followed
1	Ultimate aerobic biodegradability of plastic materials under composting conditions, Min (OR Sr. No 2)	90 percent by the end of the test period of 180 days	IS/ISO 14855-1
2	Ultimate aerobic biodegradation of plastic materials in soil, Min	90 percent of the max Level of biodegradation has been reached typically not exceeding six months but no longer than 2 years	IS/ISO 17556 CO2 evolution method
3	Regulated Heavy Metal Analysis Arsenic (as As), Cadmium (as Cd), Chromium (as Cr), Copper (as Cu), Lead (as Pb), Mercury (as Hg), Nickel (as Ni), Zinc (as Zn),	As specified in standard	IS 3025
4	Eco-toxicity Test Plant toxicity Earthworm toxicity	90 percent plant germination and Plant biomass compared to control. 90 percent of surviving earthworm grown in sample compost exposed to test material as well as reference compost exposed to reference material	OECD 208 and OECD 207

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Ultimate Aerobic Biodegradation as per ISO 17556

Breakdown of an organic compound by microorganisms in the presence of oxygen into carbon dioxide, water and mineral salts of any other elements present (mineralization) plus new biomass.

Principle:

This method is designed to yield the optimum rate of biodegradation of a plastic material in a test soil by controlling the humidity of the soil, and to determine the ultimate biodegradability of the material.

The plastic material, which is the sole source of carbon and energy, is mixed with the soil. The mixture is allowed to stand in a flask over a period of time during which the amount of oxygen consumed (BOD) or the amount of carbon dioxide evolved is determined. Provided the CO₂ evolved is absorbed, the BOD can be determined, for example, by measuring the amount of oxygen required to maintain a constant gas volume in a respirometer flask, or by measuring either automatically or manually the change in volume or pressure (or a combination of the two). The amount of carbon dioxide evolved is measured at intervals dependent on the biodegradation kinetics of the test substance by passing carbon-dioxide-free air over the soil and then determining the carbon dioxide content of the air by a suitable method.

The level of biodegradation, expressed as a percentage, is determined by comparing the BOD with the theoretical oxygen demand (ThOD) or by comparing the amount of carbon dioxide evolved with the theoretical amount (ThCO₂). The influence of possible nitrification processes on the BOD has to be considered. The normal test period is six months. The test may be shortened or extended until the plateau phase is reached, but the total test period shall not exceed two years.

Test environment

Incubation shall take place in the dark or in diffused light in an enclosure which is free from vapors toxic to microorganisms and is maintained at a temperature constant to within $\pm 2^{\circ}\text{C}$ in the range between 20°C and 28°C. preferably 25°C.

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Apparatus

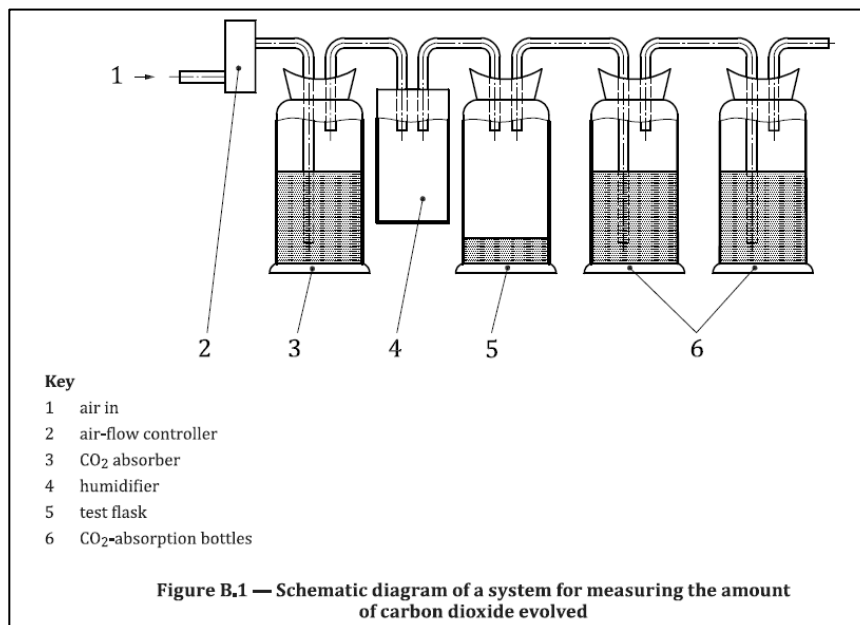


Figure 2: Apparatus for measuring the amount of carbon dioxide evolved

PROCEDURE:

Preparation of the test material

The test material shall be of known mass and contain sufficient carbon to yield a BOD or a quantity of carbon dioxide that can be adequately measured by the analytical equipment used.

The amount of test material shall be sufficient to outweigh any variations in the background oxygen consumption or any carbon dioxide evolved from the test soil: 100 mg to 300 mg of test material to 100 g to 300 g of soil is usually adequate. The maximum amount of test material is limited by the oxygen, supply to the test system. The use of 200 mg of test material with 200 g of soil is recommended unless the soil contains an excessively large amount of organic matter. When using test systems based on the determination of the carbon dioxide evolved, higher test material amounts can be used (e.g., 2500 mg for 200 g of soil) in order to increase the difference between the test material CO₂ production and

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the blank control CO₂ production. The test material should preferably be used in powder form, but it may also be introduced in the form of films, fragments or shaped articles. Test samples may be reduced in size means of cryogenic milling.

If the test material is in the form of a powder, small particles of known size distribution should be used. A particle-size distribution with its maximum at 250 um diameter is recommended. If the test material is not in powder form, the size of the pieces of material should not be greater than 5 mm x 5 mm

Preparation of the reference material

Use as reference material a well-defined biodegradable polymer (microcrystalline-cellulose powder)

Preparation of the test soil

Use natural soil collected from the surface layer of fields and/or forests. If the potential biodegradability of the test material is to be assessed, this soil may be pre-exposed to the test material. Sieve the soil to give particles of less than 5 mm, preferably less than 2 mm, in size and remove obvious plant material, stones and other inert materials. It is important to remove organic solids, such as straw, as far as practicable because they can decompose during the test and influence the results.

The soil may be pre-conditioned but, normally, pre-exposed soil should not be used, especially when biodegradation behavior in natural environments is being simulated.

Measurement of soil characteristics,

a) total water-holding capacity

b) pH of the soil

Adjustment of the water content and the pH of the soil

Adjust the water content of the soil to a suitable value for the test material by adding an appropriate amount of water to the soil, or by drying the soil in the air in a shaded place followed by addition of an appropriate amount of water. Adjust the pH of the soil to between 6.0 and 8.0 if it is not already within this range.

The optimum water content of the test soil is dependent on the test material. It is usually between

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40% and 60% of the total water-holding capacity. It is recommended that the ratio of organic carbon in the test or reference material to nitrogen in the soil (CIN ratio) be adjusted to at least 40, if it is not already at this level so as to ensure good biodegradation. This may be done by adding nitrogen as an aqueous solution of ammonium chloride or by using an aqueous solution containing the salts

Start-up and execution of the test

Prepare the following numbers of flasks:

- a) three test flasks for the test material (symbol FT);
- b) three test flasks for the blank control (symbol FB);
- c) three test flasks for checking the soil activity using a reference material (symbol Fc):

Place the soil at the bottom of each flask and add test material or reference material as indicated in the soil. Record the mass of each flask containing this test mixture. When two replicates are used, this shall be stated in the test report. It is important that the test material be homogeneously mixed with the soil, in the case of powder, and as widely spread as possible in the soil, in the case of film, to improve the contact of the test material with the microorganisms in the soil. Also, it is recommended that the surface of the test mixture be pressed with a spatula to improve the contact between the test material and the microorganisms in the soil.

Place the flasks in a constant-temperature environment and allow all the flasks to reach the desired temperature. Make all necessary connections with the respirometer or CO₂-free-air production system and start the incubation.

If measuring the carbon dioxide evolved, measure (at regular intervals depending on the carbon dioxide evolution rate) the amount of carbon dioxide evolved from each flask, using a suitable and sufficiently accurate method

When a constant level of BOD or carbon dioxide evolution is attained (plateau phase reached) and no further biodegradation is expected, the test is considered to be completed.

The test period should typically not exceed six months. However, if significant biodegradation is still observed and the plateau phase has not been reached after this length of time, then the test may be

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extended, but not to longer than two years. If the test is run for longer than six months, check periodically for possible leaks.

At the end of the test, remove the flasks and weigh them to check for any decrease in the water content of the test soil.

CALCULATIONS:

Theoretical amount of carbon dioxide evolved by test material

The theoretical amount of carbon dioxide evolved by the test material (ThCO₂) is given, in milligrams, by Formula

$$\text{ThCO}_2 = m \times w_c \times 44/12$$

where

m is the mass of test material, in milligrams, introduced into the test system.

w_c is the carbon content of the test material, determined from the chemical formula or from elemental analysis, expressed as a mass fraction:

44 and 12 are the relative molecular and atomic masses of carbon dioxide and carbon, respectively. Calculate in the same way the theoretical amount of carbon dioxide evolved by the reference material and by the mixture of test and reference material in flask FI.

Percentage biodegradation

Calculate the percentage biodegradation Dt for each test flask FT from the amount of carbon dioxide evolved during each measurement interval using Formula:

$$\frac{\Sigma m_T - \Sigma m_B}{\text{ThCO}_2} \times 100$$

ThCO₂

where

Σm_T is the amount of carbon dioxide, in milligrams, evolved in the test flask FT between the start of the test and time t;

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Σm_B is the amount of carbon dioxide, in milligrams, evolved in the blank control flask FB between the the start of the test and time t;

$ThCO_2$ is the theoretical amount of carbon dioxide, in milligrams, evolved by the test material.

RESULT:

The BIODEGRADABLE AND COMPOSTABLE THERMOFORMING PRODUCTS SAMPLE WITH 1% D2W was subjected to biodegradation as per ISO 17556 biodegradability under aerobic conditions at $25^{\circ}C \pm 2^{\circ}C$ and biodegradability was determined by measuring the actual metabolic conversion of the compostable material into carbon dioxide using the standard test method.

After 45 days of incubation under ($25^{\circ}C \pm 2^{\circ}C$), aerobic conditions using test method ISO 17556, the reference (Positive control) and BIODEGRADABLE AND COMPOSTABLE THERMOFORMING PRODUCTS SAMPLE WITH 1% D2W were gradually biodegraded.

Table 2: Percentage Biodegradation of Positive control and Test samples

Days	Percentage Biodegradation	
	Positive control	Test samples
45	61.03	2.14

----- **End of Report** -----

Authorized Signatory

Jayashree Acharya
Assistant Manager- Environment laboratory

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